

Automated Question Bank Generation System

DSA-3 Review 1

Section: A5

Team Members:

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ABOUT THE PROJECT

Introduction

The Automated Question Bank Generation System is designed to automatically generate structured examination questions from textbooks, lecture notes, and educational resources. The system extracts important keywords, classifies questions into relevant topics, estimates question difficulty, and stores them in a searchable repository. It reduces manual effort while helping educators create organized question banks efficiently.

CORE FEATURES

- 1 Keyword Extraction**
Identifies important keywords and concepts from educational content automatically.
- 2 Topic Classification**
Groups extracted content into relevant subjects and chapters.
- 3 Question Categorization**
Organizes generated questions into meaningful categories for easy access.
- 4 Difficulty Estimation**
Classifies questions as Easy, Medium, or Hard based on predefined criteria.
- 5 Searchable Repository**
Stores all generated questions in a searchable database for quick retrieval.

OUR TEAM

Team Roles & Responsibilities

1

Jaswanth

Backend Development

- Spring Boot APIs
- Business Logic
- DSA Implementation

2

Krishna

Frontend Development

- User Interface
- Dashboard Design

3

Pradeep

Database Management

- MySQL Database
- Data Integration

4

Sai Ganesh

Testing & Documentation

- Testing
- Documentation
- GitHub Repository

EXPECTED OUTCOMES

- 1 Automatically generate structured question banks
- 2 Reduce manual effort in question preparation
- 3 Categorize questions based on topics
- 4 Estimate question difficulty accurately
- 5 Maintain a searchable repository
- 6 Improve efficiency in examination preparation

WHERE IT FITS

Applications

1

Universities & Colleges

Generate organized question banks for academic assessments.

2

Schools

Support teachers in creating classroom tests and assignments.

3

EdTech Platforms

Provide automated question generation for online learning platforms.

4

Learning Management Systems (LMS)

Integrate with LMS platforms for digital assessment management.

5

Online Examination Systems

Create balanced question papers for online examinations.

6

Competitive Exam Preparation

Generate practice questions for competitive exam aspirants.

SOURCE CODE

GitHub Repository

The project source code, documentation, and future updates will be maintained in a GitHub repository.



Repository

https://github.com/jaswanthch1342/DSA-3_Project

Resources & References

RESOURCES

Programming Languages

Java, Python

Development Tools

Spring Boot, MySQL, IntelliJ IDEA / VS Code

Libraries

NLTK, spaCy

Version Control

Git, GitHub

REFERENCES

1. DSA-3 Project List (Department Project Document)
2. Oracle Java Documentation
3. Python Official Documentation
4. Spring Boot Documentation
5. NLTK Documentation
6. spaCy Documentation
7. MySQL Documentation

Thank You